

Reproduction of ggml-org/llama.cpp: Qwen2.5-7B Q4_K_M throughput and WikiText-2 perplexity

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Independent reproduction · Benchmark or leaderboard entry · Systems and performance

Abstract

Built llama.cpp at the pinned commit with CUDA for the RTX 5060 Ti (sm_120) and ran the repository's own tools on the official Qwen2.5-7B-Instruct Q4_K_M GGUF: llama-bench with its default tests (512-token prompt processing and 128-token generation, 5 repetitions, all layers on the GPU), then llama-perplexity on the WikiText-2 raw test set with default settings. One NVIDIA RTX 5060 Ti.

About this record

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	Qwen2.5-7B Q4_K_M bench ...	2026-10-07	1	1	Signed by the K-Veritas server

2 Results

r1: Qwen2.5-7B Q4_K_M bench and perplexity

Run	Metric	Final value	Step
1	pp512_tokens_per_second	3769.77	
1	pp512_stddev	103.741	
1	tg128_tokens_per_second	78.0879	
1	tg128_stddev	0.061352	
1	perplexity	0.93	
1	running_ppl	8.3223	580
1	ppl_stderr	0.06128	

Run	Inline claim	Value	Phase
1	pp512_tokens_per_second	3769.77	
1	tg128_tokens_per_second	78.0879	
1	wikitext2_ppl	8.3197	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	131.6 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	9a22d438c733523568ac8acf0a9cddcad85c2be7d796aa83ed2517e356bdb8e6	2.7 MB
r1.kvbundle.zip	d44e81d34d7ff1761a08a4ed7a7e39928e6dc88af32b46abc4b4d3e3c6c639f8	7.9 MB

5 Links and licence

Reproduction of: <https://github.com/ggml-org/llama.cpp/tree/448147d42a71d0147c6dc844f99f9ffe7419183a>

Record: CC-BY 4.0. Code: MIT, as declared by the authors. Tags: llama.cpp, inference, quantization, perplexity, benchmark.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (Qwen2.5-7B Q4_K_M bench and perplexity), kv:2610.00015v1/r1. Data hash b984556d40a3984d6689ad4afa528098bf4626b887cada3a059b5bfa54876972.

<https://kveritas.org/records/2610.00015v1/r1.pdf>